

Self-Learning: Red-Black Trees Data Structures

> **Definition:** A **Red-Black Tree** is a self-balancing Binary Search Tree where each node stores an extra bit representing **color** (Red or Black), satisfying specific balance constraints that ensure the tree height remains bounded by $2 \log_2(N + 1)$.

1. Detailed Technical Explanation

The 5 Red-Black Tree Properties:

1. **Node Color:** Every node is either **RED** or **BLACK**.

2. **Root**

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2. AVL Tree vs Red-Black Tree Comparison

| Feature | AVL Tree | Red-Black Tree |

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3. Insertion in Red-Black Tree

When inserting a new key:

1. Inse

4. Quick Recall Flow

RB-Tre